Lending Under Imperfect Information

November 2 follow-up: two scenarios and three methods

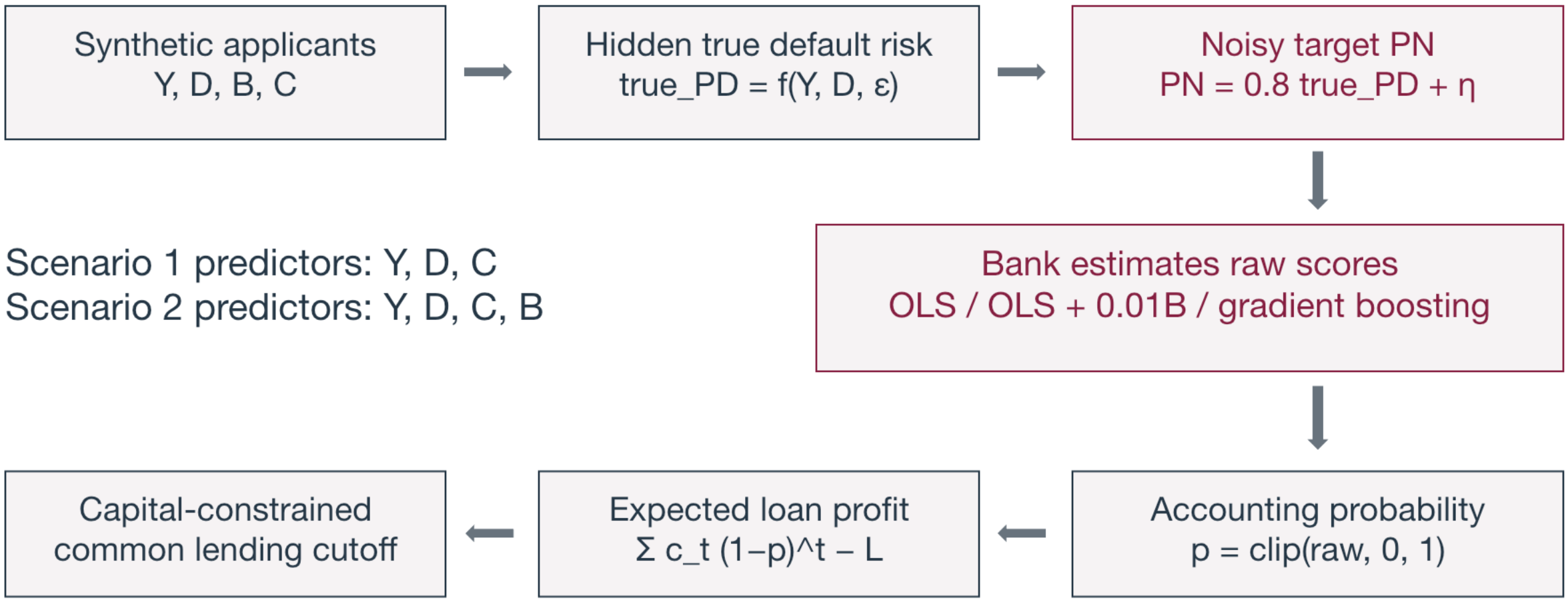
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2 November 2026

Corrected +0.01B penalty
Systemic risk and gradient-boosting comparison

100,000 applicants. 80,000 training. 20,000 evaluation.

Single matched synthetic run. Earlier studies appear only in the historical appendix.

From True Risk to Lending Decisions



Scenario 1 predictors: Y, D, C
Scenario 2 predictors: Y, D, C, B

Approval decisions, group outcomes, and evaluator-only true expected profit

Working contract: T=30, theta=0.10, r=5% per period. Fixed capital $1.826087B.

Scenario 1: No Systemic Differences

B=0 and B=1 share the same underlying risk process

The bank fits PN using income, DTI and the credit signal.

Only the manual-penalty branch explicitly adds +0.01B.

Finite-sample group differences can still appear without a group effect in true risk.

Original group-neutral population. Same applicants, saved noise and fixed evaluation cohort.

Scenario 1: Methods and Corrected Penalty

Higher estimated default risk disfavors B=1

Method	Raw score	Information / adjustment
OLS control	f_OLS(Y, D, C)	Unadjusted fitted score
OLS + penalty	f_OLS(Y, D, C) + 0.01 B	Same OLS fit, +1 pp for B=1
Gradient boosting	f_ML(Y, D, C)	Same predictors, no added penalty

The control and penalty share exactly the same OLS fit

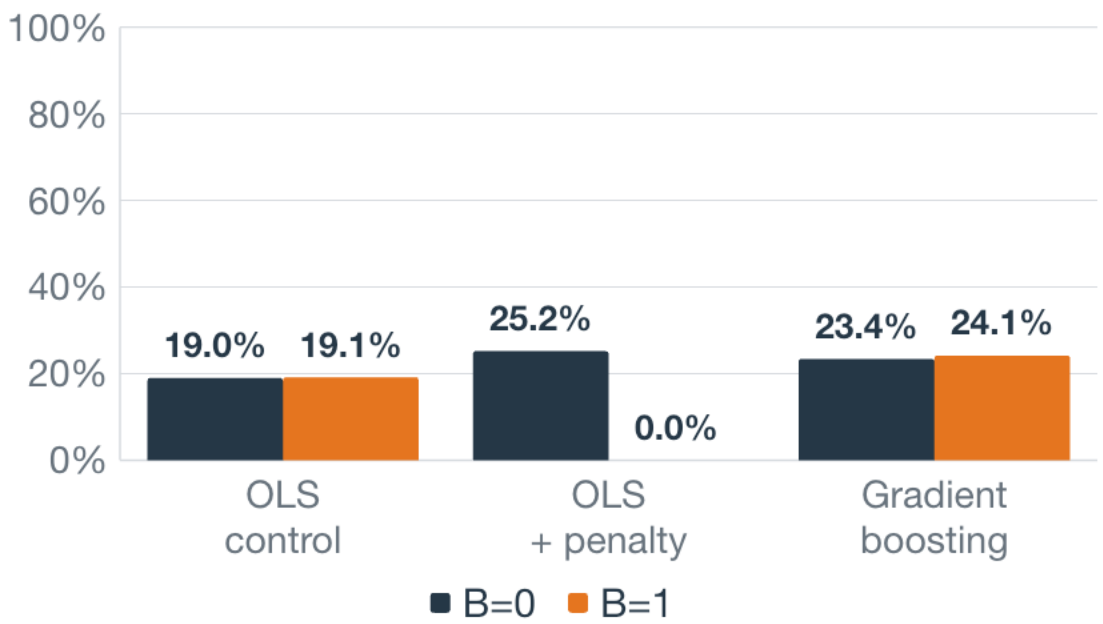
ML settings use training-only validation PN error. Evaluation truth stays hidden.

PN remains the continuous regression target. Raw scores and bounded accounting PD remain separate.

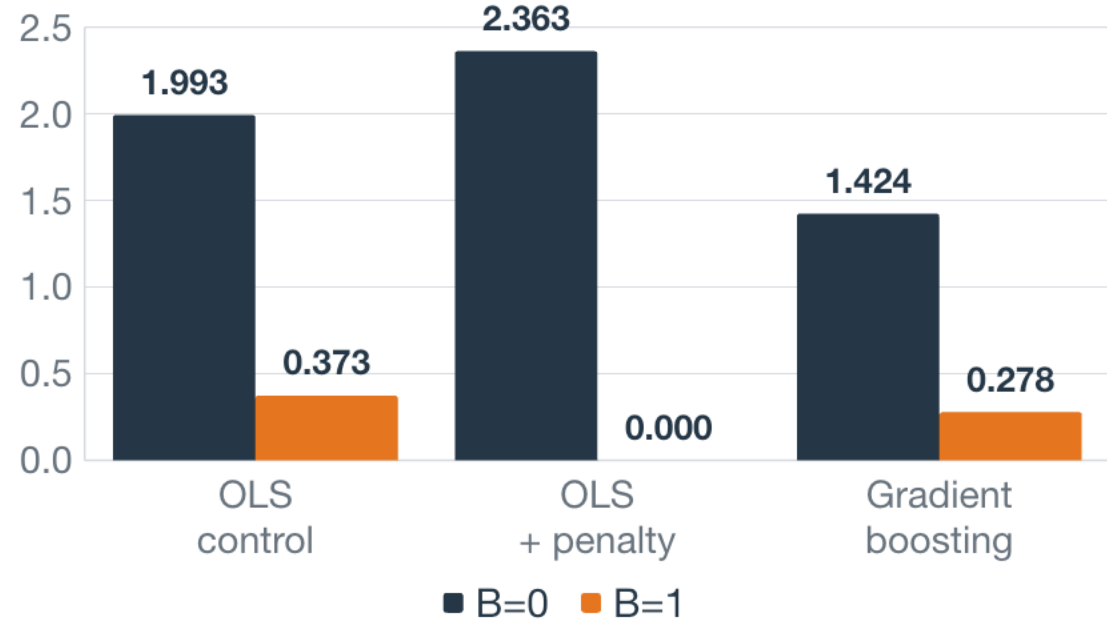
Scenario 1: Approval Rates and True Profit

ML uses 72.4% of capital. The next whole score group exceeds budget

Within-group approval rate



True expected profit by group ($B)



Penalty minus control: true portfolio profit –$3.821M

Profit bars sum funded-loan contributions. Group sizes: B=0 16,970 and B=1 3,030.

Source: nov2_followup/group_results.csv and portfolio_results.csv. Same contracts and capital.

Scenario 2: Systemic Differences

B=1 has a higher hidden risk factor

$$\varepsilon_2 = \varepsilon_1 + B$$

$$C_2 = 0.4\ \varepsilon_2 + \omega$$

$PN_2 = 0.8$ true_PD_2 + the same η

OLS and ML now use Y, D, C and B

B=1 population mean true PD: 0.14229% to 0.14476%.
Between-world results also reflect the change in permitted predictors.

Researcher-chosen +1 SD epsilon shift for B=1. Frozen true-risk coefficients remain unchanged.

Scenario 2: Methods and Corrected Penalty

Higher estimated default risk disfavors B=1

Method	Raw score	Information / adjustment
OLS control	f_OLS(Y, D, C, B)	Unadjusted fitted score
OLS + penalty	f_OLS(Y, D, C, B) + 0.01 B	Same OLS fit, +1 pp for B=1
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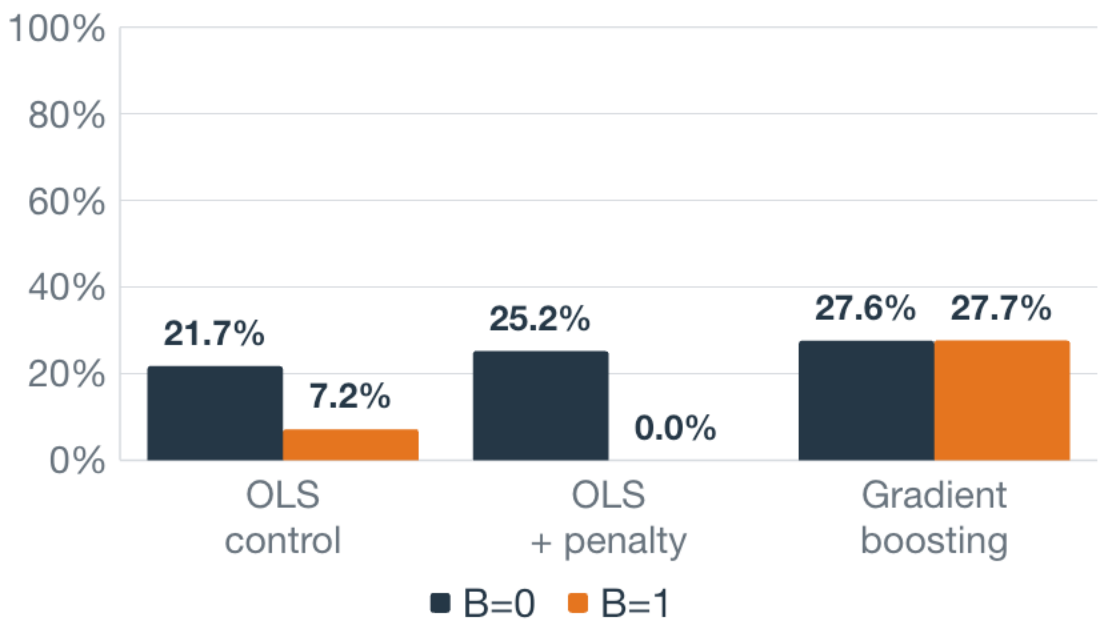
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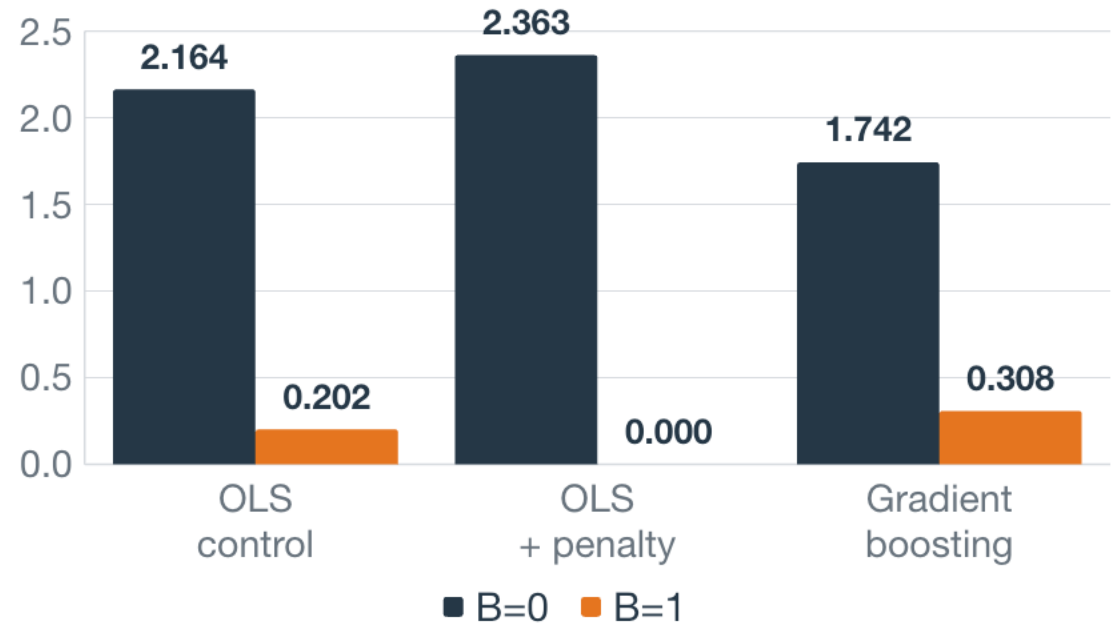
Scenario 2: Approval Rates and True Profit

ML uses 87.2% of capital. The next whole score group exceeds budget

Within-group approval rate



True expected profit by group ($B)



Penalty minus control: true portfolio profit –$3.179M

Profit bars sum funded-loan contributions. Group sizes: B=0 16,970 and B=1 3,030.

Source: nov2_followup/group_results.csv and portfolio_results.csv. Same contracts and capital.

Moving the Tie Leaves Lending Unchanged

Replace only exact zeros with a predeclared positive floor

Quantity	Original direct policy	Positive-floor sensitivity
Shared tie score	0%	0.013924%
Applicants in the tie	3,764	3,764
Total approvals	4,990	4,990
True portfolio profit	$2.3593B	$2.3593B

0 approval decisions change

Perceived profit changes by –$7.855M. The tie remains intact.

Source: mechanism_audit/positive_floor_derivation.json and portfolio_comparison.csv. Canonical sensitivity.

Removing B's Contribution Without Refitting

raw_Z = raw_W – saved_beta_B × B

W keeps B. Z removes only its contribution. F is the full blind refit.

Approval gap, B=1 minus B=0 (pp)



Z minus W true profit

+$853,049

Z minus F true profit

+$327,762

Z and F have equal group counts but exchange one approved loan.

Source: fixed_coefficient/cd7f3ee268cfc4d0/canonical_policies.csv. One canonical realization.

No Universal Profit Gain Across 200 Draws

Share of draws by signed true-profit difference



Comparison	Mean change	Median change	MCSE of mean
Z minus W	+$12.82M	+$0.593M	$20.53M
Z minus F	–$11.80M	$0	$26.50M

Rare reject-all disagreements dominate the means

Source: fixed_coefficient/cd7f3ee268cfc4d0/noise_summary.csv. Same population and 200 paired saved noise draws.

Ranking Scores and Accounting Probabilities

Role	Original bounded policy	Raw-order sensitivity
Lending order	clip(raw, 0, 1)	raw score
Cash-flow PD	clip(raw, 0, 1)	Same clip(raw, 0, 1)
Indivisible ties	Exact bounded scores	Exact raw scores

Draw 008, aware: the zero-score block requires 110.01% of capital.

Raw ordering permits a feasible prefix within the clipped block

Every evaluation request is perceived as profitable, so capital limits lending.
Raw scores rank requests. Cash flows use the same bounded PD.

Source: raw_ordering.yaml and saved raw-ordering bank freezes. Fixed probabilities, contracts and capital.

Nine Reject-All Policies Now Fund Loans

Approvals shown as original / raw ordering

Case	W: aware	Z: B removed	F: blind refit
Canonical	3,900 / 3,900	3,802 / 3,802	3,802 / 3,802
Draw 008	0 / 5,107	0 / 4,741	4,744 / 4,744
Draw 064	0 / 3,898	3,701 / 3,701	0 / 3,701
Draw 089	4,331 / 4,331	0 / 4,163	4,161 / 4,161
Draw 092	0 / 4,222	0 / 3,865	3,861 / 3,861
Draw 154	0 / 4,812	3,454 / 3,454	0 / 3,454

Risk estimates and per-request profits stay unchanged

The aware approval gap in draw 154 remains +88.25 pp.

Source: raw_ordering/d0421da2ea691f2e/policies.csv. Six selected diagnostics, not a random uncertainty sample.

Decisions Before the Final Research Report

Lending order	May raw scores rank requests while bounded PD determines profits?
Contract interpretation	Is 5% per period with the 30-period growing schedule intended?
Measurement noise	Keep SD 0.1 as the working case, or predeclare a separate noise study?
Paper scope	Should the final paper add an ML benchmark for this model?

Next deliverable: a final analysis protocol and paper outline